

CE 49X: Introduction to Computational Thinking and Data Science

Midterm Exam – Part 1

— ANSWER KEY —

Section A: Multiple Choice [20 points]

1. (b) 4 — `range(10)` produces 0–9; values divisible by 3: $\{0, 3, 6, 9\} = 4$ elements.
2. (c) `import numpy as np` — This creates the alias `np` for the `numpy` module.
3. (b) `array([2, 3, 4])` — Slicing `a[1:4]` returns elements at indices 1, 2, 3 (end index is exclusive).
4. (b) `df.fillna(method='ffill')` — Forward fill propagates the last valid value forward. `dropna()` removes rows; `interpolate()` computes intermediate values; `fillna(0)` fills with zero.
5. (d) **Position** — Position on a common scale is the most accurate visual encoding channel for quantitative comparison (Cleveland & McGill hierarchy).
6. (b) **Data-ink ratio** — Tufte’s principle: maximize the proportion of ink used to display data vs. non-data elements (gridlines, decoration, chart junk).
7. (b) 2.0 — $z = (66 - 50)/8 = 16/8 = 2.0$.
8. (b) **Reject H_0** — Since $p = 0.03 < \alpha = 0.05$, we reject the null hypothesis. Note: (a) is a common misconception; p-value is NOT the probability that H_0 is true.
9. (b) **It never identifies the dangerous 5%** — This is the “accuracy trap.” A model that always predicts the majority class achieves high accuracy but has zero recall for the minority class. It is useless for detecting the cases that matter.
10. (b) **Features are conditionally independent given the class** — The “naive” assumption is that $P(f_1, f_2, \dots, f_n|L) = \prod_i P(f_i|L)$. This simplifies computation but is rarely true in practice.

Section B: Written Questions [30 points]

Question 1: Code Tracing & Debugging [6 points]

(a) (3 points)

Answer

The list comprehension filters beams > 5.0 : `[7.2, 12.1, 6.3]`.

`total = 7.2 + 12.1 + 6.3 = 25.6`, `count = 3`.

Output:

```
Found 3 beams totaling 25.6 m
Average: 8.53 m
```

Grading: 1 pt for correct list comprehension result, 1 pt for first line, 1 pt for second line with correct formatting.

(b) (3 points)

Answer

Bug: The function uses the SPT-N value `n` as the dictionary key. When duplicate values appear in the input (5 appears twice, 25 appears twice), the second occurrence overwrites the first. The dictionary ends up with only 3 unique keys: `{5: 'soft', 25: 'medium', 40: 'hard'}`.

Current output: 3 (instead of the expected 5).

Fix: Use a list instead of a dict, or use `enumerate` to create unique keys:

```
results = []                # Use a list instead
for n in spt_n_values:
    if n < 10:
        results.append('soft')
    ...
```

Grading: 1 pt for identifying the bug (dict keys overwrite), 1 pt for stating current output is 3, 1 pt for a valid fix.

Question 2: NumPy & Pandas [6 points]

(a) (3 points)

Answer

A **view** shares memory with the original array — modifications to the view affect the original. A **copy** is an independent array with its own memory.

NumPy slicing creates a **view**. So `subset = speeds[1:4]` is a view of `speeds[1]`, `speeds[2]`, `speeds[3]`. Setting `subset[0] = 99` modifies `speeds[1]`.

Answer: `speeds[1]` is **99**.

Grading: 1 pt for explaining view vs copy, 1 pt for stating slicing creates a view, 1 pt for correct answer (99).

(b) (3 points)

Answer

```
df[(df['AVERAGE_SPEED'] < 20) & (df['NUMBER_OF_VEHICLES'] > 100)][['GEOHASH', 'AVERAGE_SPEED']]
```

Also acceptable with `.loc`:

```
df.loc[(df['AVERAGE_SPEED'] < 20) & (df['NUMBER_OF_VEHICLES'] > 100), ['GEOHASH', 'AVERAGE_SPEED']]
```

Grading: 1 pt for correct boolean conditions with `&`, 1 pt for parentheses around each condition, 1 pt for correct column selection. Deduct 1 pt if using `and` instead of `&` (won't work on Series).

Question 3: Visualization [6 points]**(a) (3 points)****Answer**

Problem 1: A pie chart with 8 categories makes angular comparison nearly impossible. The 3D perspective further distorts proportions (slices closer to the viewer appear larger). **Fix:** Use a sorted horizontal bar chart — position/length encoding is far more accurate than angle.

Problem 2: The jet (rainbow) colormap has no perceptual ordering — it creates false boundaries and is not colorblind-friendly. **Fix:** Use a sequential colormap like `YlOrRd` or `viridis` where darker/brighter corresponds to higher values.

Grading: 1.5 pts per problem identified + fix. Accept any two valid issues from: too many slices for pie, 3D distortion, rainbow colormap. Accept reasonable alternative fixes.

(b) (3 points)**Answer**

Anscombe's Quartet consists of four datasets that have nearly identical summary statistics (same mean, standard deviation, correlation coefficient, and linear regression line) but look completely different when plotted.

Lesson: Summary statistics alone can hide critical patterns in data. One dataset is linear, one is curved, one has a single outlier pulling the regression, and one has a cluster with an outlier.

Always visualize your data before drawing conclusions or building models.

Grading: 1 pt for correct definition (4 datasets, same statistics), 1 pt for stating they look different visually, 1 pt for the lesson (must visualize, statistics can mislead).

Question 4: Statistical Analysis [6 points]**(a) (3 points)****Answer**

$$z = \frac{x - \mu}{\sigma} = \frac{26.1 - 32.5}{3.2} = \frac{-6.4}{3.2} = -2.0$$

$|z| = 2.0$. By strict inequality $|z| > 2$, this is exactly at the boundary and **not** extreme. By $|z| \geq 2$, it **is** extreme.

Grading: 2 pts for correct z-score calculation, 1 pt for correct classification with justification. Accept either interpretation if justified.

(b) (3 points)

Answer

Type I error (false positive): Rejecting H_0 when it is true. Here: concluding the concrete **does not** meet the 30 MPa specification when it actually does. Result: a good concrete batch is rejected, wasting money and causing delays.

Type II error (false negative): Failing to reject H_0 when it is false. Here: concluding the concrete **meets** the specification when it actually does not. Result: substandard concrete is used in construction.

Type II is more dangerous for structural safety because it leads to using weak concrete in structures, risking collapse and endangering lives. Type I only wastes resources.

Grading: 1 pt for each correct definition with engineering interpretation, 1 pt for identifying Type II as more dangerous with justification.

Question 5: Machine Learning & Naive Bayes [6 points]**(a) (3 points)****Answer**

For the “Liquefiable” class: $TP = 20$, $FP = 15$, $FN = 5$.

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{20}{20 + 15} = \frac{20}{35} \approx 0.571$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{20}{20 + 5} = \frac{20}{25} = 0.80$$

Prioritize recall. Missing a liquefiable site (false negative) means building on dangerous ground, risking structural collapse during an earthquake. A false positive only triggers unnecessary ground improvement — costly but not life-threatening.

Grading: 1 pt for correct precision, 1 pt for correct recall, 1 pt for correct prioritization with engineering justification.

(b) (3 points)**Answer**

Using Bayes’ theorem:

$$P(\text{Poor} | 8\text{mm}) = \frac{P(8\text{mm} | \text{Poor}) \cdot P(\text{Poor})}{P(8\text{mm})}$$

First, compute $P(8\text{mm})$ using the Law of Total Probability:

$$\begin{aligned} P(8\text{mm}) &= P(8\text{mm} | \text{Good}) \cdot P(\text{Good}) + P(8\text{mm} | \text{Poor}) \cdot P(\text{Poor}) \\ &= (0.02)(0.70) + (0.15)(0.30) \\ &= 0.014 + 0.045 = 0.059 \end{aligned}$$

Then:

$$P(\text{Poor} | 8\text{mm}) = \frac{(0.15)(0.30)}{0.059} = \frac{0.045}{0.059} \approx \boxed{0.763}$$

Despite the prior belief that only 30% of beams are in poor condition, observing an 8mm crack raises this to $\approx 76\%$.

Grading: 1 pt for correct $P(8\text{mm})$ via LOTP, 1 pt for correct application of Bayes’ formula, 1 pt for correct final answer (≈ 0.76).

End of Answer Key